ORF 445 High Frequency Markets
Homework 5

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April 14, 2026
due midnight Sunday, April 26

1. Measure market impact following an aggressive order. Do this for one symbol of your choice, averaging over at least three weeks of data. The result will be a curve of average midpoint price change for different lags following a market order.

   (a) As in class, measure impact following a market order for up to 50 trades after. Take the trade sign from the CME `aggr` variable, and ignore trades that have a null value of `aggr`. (Trades with null `aggr` still count for lag and price change, but are not part of the data sample for averaging.) Do only the step function ("constant") model, where the impact depends only on the sign of the trade, not its size. That is, you simply take the average forward price change at each lag, with positive sign for buy orders, negative for sell orders, and zero for null sign.

   (b) Evaluate the importance of trade size by generating two curves, one curve for market orders whose size is smaller than the median size, and another one for market orders whose size is larger than the median. Do your results support the hypothesis that large trades have more impact than small trades?

   (c) Again generate two subsets and two curves, this time separating by quote size on the side of the market to which the aggressive trade is sent. Do your results support the hypothesis that trades sent into deeper markets (larger quote size) have smaller impact?

2. Pick a reasonably active symbol such as the SP500 futures `ESH4`. Build an intraday liquidity curve in the following way:

   (a) For each minute on each date (in some reasonable range, ideally the whole month), extract 1-minute bid-ask midpoint price samples p, using `aj[]`, and take the differences Δp.

   (b) For each minute, extract the net traded volume v: buy volume v_B minus sell volume v_S. For a product like `ESH4` that has a small amount of implied trading, it will be enough to use the `aggr` tag; for an energy or agricultural product you will need to match the bid and ask.

   (c) For any particular time period, do a linear regression of Δp against v, with zero intercept so $\Delta p = \lambda v$. (Be sure that the price change and the traded volume are for the same minute, not off by one.) The coefficient λ is an illiquidity as in Kyle's model: how much the price changes per unit net volume traded. For this problem we neglect refinements, such as taking account of the average quote size for different trade volumes.

   (d) Do this for each hour in the trading day, across your entire date range: -07:00 to -06:00, -06:00 to -05:00, …, 15:00 to 16:00. The result will be a curve of estimated market impact through the day.